

We are given triangle ABC with side lengths $AB = 26$, $BC = 17$, $AC = 25$. We set up a coordinate system with $B = (0, 0)$, $C = (17, 0)$, and solve for A . From the distances we find $A = (10, 24)$. The orthocenter H is the intersection of the altitudes; since BC is horizontal, the altitude from A is the vertical line $x = 10$. The altitude from B is perpendicular to AC ; its equation is $24t = 10$ giving $t = \frac{5}{12}$, so $H = (10, \frac{35}{12})$.

The feet of the altitudes are

$$D = (10, 0), \quad E = \left(\frac{9792}{625}, \frac{2856}{625}\right), \quad F = \left(\frac{425}{169}, \frac{1020}{169}\right).$$

Line EF meets BC (the x -axis) at X . Using the two-point form we obtain

$$X = \left(\frac{170}{3}, 0\right).$$

The circumcircle Ω has centre $O = (\frac{17}{2}, \frac{253}{24})$ and radius $R = \frac{325}{24}$.

Line AX is parametrised as $A + s(X - A)$. Substituting into the equation of Ω gives a quadratic in s . Besides $s = 0$ (point A) we obtain

$$s = \frac{759}{12392}.$$

Hence the second intersection Q of AX with Ω is

$$Q = \left(10 + 140 \cdot \frac{759}{12392}, 24 - 72 \cdot \frac{759}{12392}\right) = \left(\frac{57545}{3098}, \frac{30345}{1549}\right).$$

The nine-point circle (circle DEF) has centre $N = (\frac{37}{4}, \frac{323}{48})$ and radius $R_N = \frac{325}{48}$. A well-known homothety with centre H and factor $\frac{1}{2}$ sends the circumcircle to the nine-point circle. Therefore the image of Q under this homothety lies on the nine-point circle and on the segment HQ . That image is exactly the midpoint of H and Q ; call it P . Thus

$$P = \frac{H + Q}{2}.$$

Because P is the midpoint of HQ , we have $PQ = \frac{1}{2}HQ$. The required ratio is

$$\frac{PQ}{PE} = \frac{HQ}{2PE}.$$

We now compute HQ^2 and PE^2 using the coordinates. Let $d = 1549$ (note that $3098 = 2d$). Then

$$Q = \left(\frac{57545}{2d}, \frac{30345}{d}\right), \quad H = \left(10, \frac{35}{12}\right).$$

$$\Delta x = Q_x - H_x = \frac{57545 - 20d}{2d}, \quad \Delta y = Q_y - H_y = \frac{364140 - 35d}{12d}.$$

With $d = 1549$ we get $57545 - 20d = 26565$ and $364140 - 35d = 309925$. Hence

$$HQ^2 = \frac{26565^2}{4d^2} + \frac{309925^2}{144d^2} = \frac{36 \cdot 26565^2 + 309925^2}{144d^2}.$$

Evaluating the squares:

$$\begin{aligned} 26565^2 &= 705\,699\,225, & 309925^2 &= 96\,053\,505\,625, \\ 36 \cdot 26565^2 &= 25\,405\,172\,100, & \text{sum} &= 121\,458\,677\,725. \end{aligned}$$

Thus

$$HQ^2 = \frac{121\,458\,677\,725}{144\,d^2}.$$

Factorising the numerator shows it is $1549 \times 78\,411\,025$. Since $d = 1549$, we obtain

$$HQ^2 = \frac{78\,411\,025}{144\,d}.$$

But $78\,411\,025 = 8855^2$ (because $8854^2 = 78\,393\,316$ and $8855^2 = 78\,393\,316 + 17\,709 = 78\,411\,025$). Therefore

$$HQ^2 = \frac{8855^2}{144\,d} \implies PQ^2 = \frac{HQ^2}{4} = \frac{8855^2}{576\,d}.$$

Now we compute PE^2 . Because P and E lie on the nine-point circle, we use the relation

$$PE^2 = 2R_N^2 - 2\overrightarrow{NP} \cdot \overrightarrow{NE}.$$

In the scaled system (multiplied by 48 to clear denominators) we have

$$N' = (444, 323), \quad P' = \left(240 + \frac{690540}{d}, 70 + \frac{728280}{d}\right), \quad E' = \left(\frac{470016}{625}, \frac{137088}{625}\right).$$

The dot product $\overrightarrow{NP} \cdot \overrightarrow{NE}$ in the original coordinates is

$$\overrightarrow{NP} \cdot \overrightarrow{NE} = \frac{50\,312\,467\,283}{625\,d \cdot 2304}.$$

The squared radius is $R_N^2 = \left(\frac{325}{48}\right)^2 = \frac{105625}{2304}$. Hence

$$PE^2 = 2 \cdot \frac{105625}{2304} - 2 \cdot \frac{50\,312\,467\,283}{625\,d \cdot 2304} = \frac{211250}{2304} - \frac{100\,624\,934\,566}{625\,d \cdot 2304}.$$

Writing the first term with denominator $625d \cdot 2304$ gives

$$PE^2 = \frac{211250 \cdot 625d - 100\,624\,934\,566}{625d \cdot 2304}.$$

Compute $211250 \cdot 625 = 132\,031\,250$. Then

$$211250 \cdot 625d = 132\,031\,250 \cdot 1549 = 204\,516\,406\,250,$$

and subtracting $100\,624\,934\,566$ yields $103\,891\,471\,684$. Therefore

$$PE^2 = \frac{103\,891\,471\,684}{625d \cdot 2304}.$$

Factorising the numerator: $103\,891\,471\,684 = 4 \cdot 25\,972\,867\,921$. Notice that $25\,972\,867\,921 = 161?$ Actually we observe that

$$103\,891\,471\,684 = 8855^2 \cdot 625d.$$

Indeed, $8855^2 = 78\,411\,025$ and

$$78\,411\,025 \times 625d = 78\,411\,025 \times 625 \times 1549.$$

Compute $78\,411\,025 \times 625 = 49\,013\,140\,625$ (since $78\,411\,025 \times 600 = 47\,046\,615\,000$ and $\times 25 = 1\,960\,275\,625$, sum $49\,006\,890\,625$? Let's do carefully:

$$78\,411\,025 \times 625 = 78\,411\,025 \times \left(\frac{5^4}{1}\right) = \frac{78\,411\,025 \times 5^4}{1}.$$

But it is easier to verify the equality by checking that

$$\frac{103\,891\,471\,684}{625d} = 8855^2.$$

Compute $103\,891\,471\,684 \div (625d) = \frac{103\,891\,471\,684}{625 \times 1549}$. Since $103\,891\,471\,684 = 8855^2 \times 625d$ is exactly what we need, we can confirm by noting that

$$8855^2 \times 625d = 78\,411\,025 \times 625 \times 1549.$$

Now $78\,411\,025 \times 625 = 49\,013\,140\,625$ (multiply by 5^4 : $78\,411\,025 \times 5 = 392\,055\,125$, $\times 5 = 1\,960\,275\,625$, $\times 5 = 9\,801\,378\,125$, $\times 5 = 49\,006\,890\,625$? Wait, we need to be systematic:

$$78\,411\,025 \times 5 = 392\,055\,125,$$

$$392\,055\,125 \times 5 = 1\,960\,275\,625,$$

$$1\,960\,275\,625 \times 5 = 9\,801\,378\,125,$$

$$9\,801\,378\,125 \times 5 = 49\,006\,890\,625.$$

So $78\,411\,025 \times 625 = 49\,006\,890\,625$. Then

$$49\,006\,890\,625 \times 1549 = 49\,006\,890\,625 \times (1500 + 49) = 73\,510\,335\,937\,500 + 2\,401\,337\,640\,625 = 75\,911\,673\,578\,125.$$

That is far larger than $103\,891\,471\,684$. Something is off. Let's recalc: Actually $625d = 625 \times 1549 = 968\,125$. So we need to check whether

$$8855^2 \times 968\,125 = 103\,891\,471\,684.$$

Compute $8855^2 = 78\,411\,025$. Multiply by $968\,125$:

$$78\,411\,025 \times 968\,125.$$

We can compute $78\,411\,025 \times 968\,125 = 78\,411\,025 \times (1\,000\,000 - 31\,875)$? That seems messy. Instead, note that earlier we derived

$$PE^2 = \frac{103\,891\,471\,684}{625d \cdot 2304}.$$

We also have $PQ^2 = \frac{8855^2}{576d}$. Therefore

$$\frac{PQ^2}{PE^2} = \frac{8855^2 / (576d)}{103\,891\,471\,684 / (625d \cdot 2304)} = \frac{8855^2}{576d} \cdot \frac{625d \cdot 2304}{103\,891\,471\,684} = \frac{8855^2 \cdot 625 \cdot 2304}{576 \cdot 103\,891\,471\,684}.$$

Simplify $2304/576 = 4$. Hence

$$\frac{PQ^2}{PE^2} = \frac{8855^2 \cdot 625 \cdot 4}{103\,891\,471\,684} = \frac{8855^2 \cdot 2500}{103\,891\,471\,684}.$$

Now compute $103\,891\,471\,684 \div 2500 = 41\,556\,588.6736$, not an integer. But we suspect that $103\,891\,471\,684 = 8855^2 \times 2500$? Let's test: $8855^2 = 78\,411\,025$. Multiply by $2500 = 78\,411\,025 \times 2500 = 78\,411\,025 \times 1000 \times 2.5 = 78\,411\,025\,000 \times 2.5 = 196\,027\,562\,500$. That is far larger. So my earlier guess that numerator equals $8855^2 \times 625d$ is incorrect.

We need to find the correct factorisation. Let's compute $103\,891\,471\,684$ divided by $d = 1549$. $1549 \times 67\,000\,000 = 103\,783\,000\,000$, remainder $108\,471\,684$. $1549 \times 70\,000 = 108\,430\,000$, remainder $41\,684$. $1549 \times 27 = 41\,823$, too high. So not a simple multiple.

Maybe we made an arithmetic mistake in the dot product or in the subtraction. Let's recompute the dot product more carefully.

We had

$$\overrightarrow{NP} \cdot \overrightarrow{NE} = \frac{50\,312\,467\,283}{625d \cdot 2304}.$$

But we should verify this number. It came from a lengthy computation that might contain errors. Instead of trusting that, we can compute PE^2 directly from the coordinates of P and E using the simplified expressions we found for Δx and Δy .

Recall we obtained

$$\Delta x = P_x - E_x = -\frac{5\,318\,107}{2500d}, \quad \Delta y = P_y - E_y = \frac{155\,347\,219}{15000d}.$$

But note that the denominators are different: $2500d$ and $15000d$. To combine them for the squared distance we need a common denominator. The least common multiple of 2500 and 15000 is 15000 . So rewrite Δx with denominator $15000d$:

$$\Delta x = -\frac{5\,318\,107}{2500d} = -\frac{5\,318\,107 \times 6}{15000d} = -\frac{31\,908\,642}{15000d}.$$

Indeed $5\,318\,107 \times 6 = 31\,908\,642$. And $\Delta y = \frac{155\,347\,219}{15000d}$. Therefore

$$PE^2 = \frac{31\,908\,642^2 + 155\,347\,219^2}{(15000d)^2}.$$

Now compute the squares. Let $A = 31\,908\,642$, $B = 155\,347\,219$. We need $A^2 + B^2$. Notice that $B = 5 \times A + ?$ $A \times 5 = 159\,543\,210$, not B . Perhaps A and B are multiples of 8855 ? $8855 \times 3600 = 31\,898\,000$, close to A . $8855 \times 3601 = 31\,907\,855$, difference 787 . Not exact.

Given the enormous numbers, it is likely that $A^2 + B^2$ simplifies to a multiple of d^2 times a perfect square, leading to a nice ratio. However, the algebra is extremely heavy. Perhaps there is a much simpler synthetic approach that yields the ratio directly.

We already know that P is the midpoint of HQ . Maybe we can compute HQ and PE using known lengths in the triangle. Let's try to find coordinates for Q in a different way.

From the homothety, Q is the point on the circumcircle such that P is the midpoint of HQ . Since P lies on the nine-point circle, Q is the reflection of H across P . But P is also the midpoint of HQ , so Q is the reflection of H across P . Not helpful.

Maybe we can use complex numbers with the circumcircle as unit circle. Set the circumradius to 1 by scaling the triangle. The side lengths become

$$a' = \frac{17}{R}, \quad b' = \frac{25}{R}, \quad c' = \frac{26}{R},$$

where $R = \frac{325}{24}$. So $a' = \frac{17 \cdot 24}{325} = \frac{408}{325}$, $b' = \frac{600}{325} = \frac{24}{13}$, $c' = \frac{624}{325}$. These are rational. Place the circumcenter at the origin. Let the vertices be complex numbers a, b, c on the unit circle. Then $h = a + b + c$. The nine-point circle has centre $n = h/2$ and radius $1/2$. The point E (foot from B to AC) can be expressed as

$$e = \frac{a + c - b + ac\bar{b}}{2}?$$

I recall a formula: In a triangle with circumcenter 0, the foot of the altitude from B to AC is

$$e = \frac{a + c - ac\bar{b}}{2}.$$

Let's verify: The line AC has equation $z + ac\bar{z} = a + c$. The altitude from B is perpendicular to AC , so it satisfies $(z - b) \perp (c - a)$. One can derive that the foot is $e = \frac{a+c-ac\bar{b}}{2}$. Indeed, this is a known formula. Similarly, the